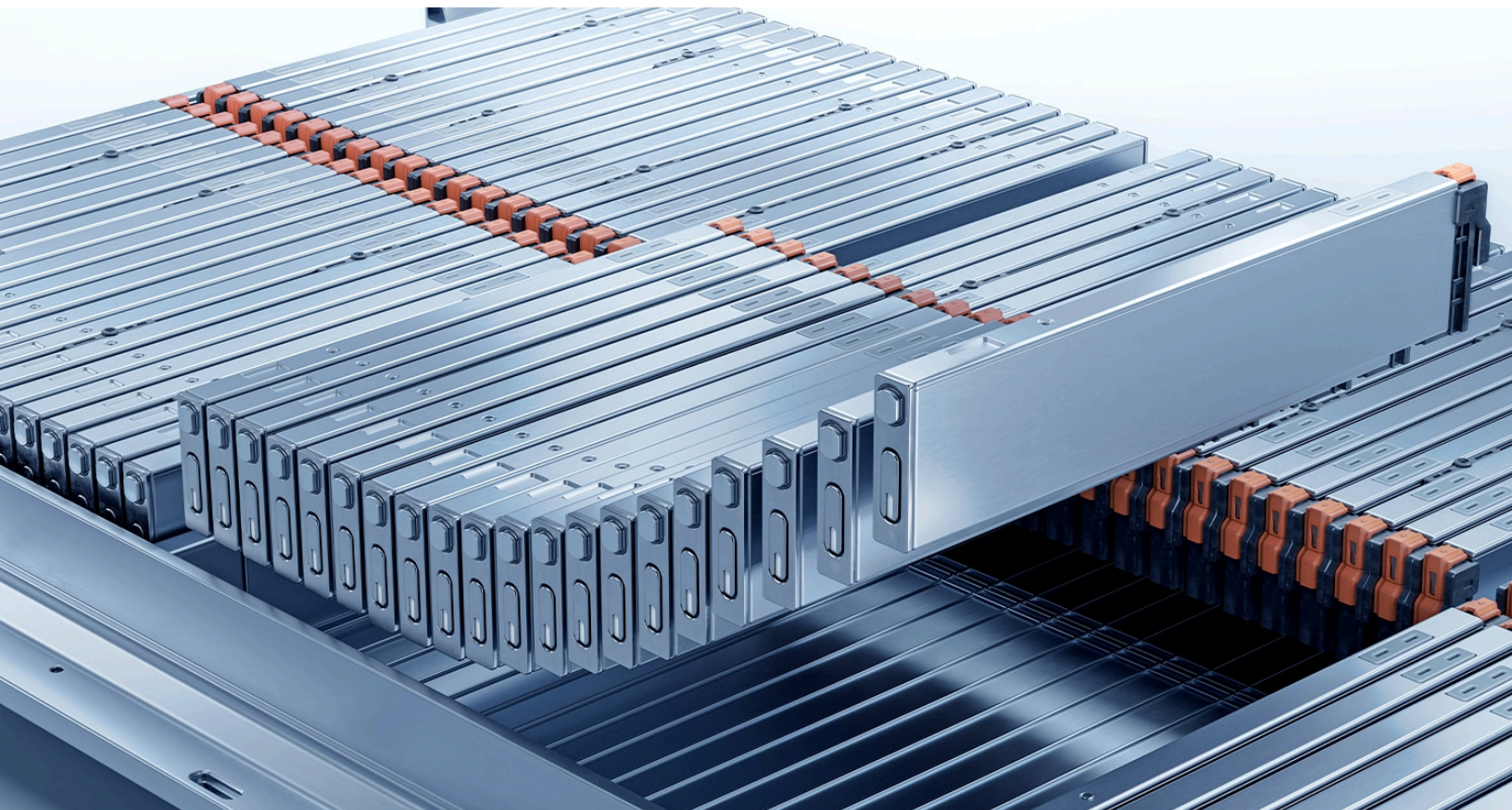


Automotive & Assembly Practice

Battery 2035: Building new advantages

Lowering costs and improving performance for battery electric vehicles entails adopting rapidly evolving technologies, localizing supply chains, and overcoming production bottlenecks in the battery industry.

This article is a collaborative effort by Andreas Breiter, Denny Thiemig, Patrick Schaufuss, Raphael Rettig, and Wenting Gao, with Yunjing Kinzel, representing views from McKinsey's Battery Accelerator Team.



The global lithium-ion (Li-ion) battery industry finds itself at a new inflection point. Demand for Li-ion batteries crossed the milestone threshold of 1.0 terawatt-hours (TWh) in 2024 and likely reached nearly 1.6 TWh in 2025.¹ Global installations of battery energy storage systems (BESS) are projected to hit annual records through 2035.² Yet pack prices in 2024 saw their largest annual drop in more than five years, and in 2025, they fell further to a record low of \$108 per kilowatt-hour (kWh)—less than half the price in 2018—with cell prices dropping to \$74 per kWh.³ This was primarily due to overcapacity in battery cell manufacturing.

Global overcapacity of Li-ion batteries was approximately 900 gigawatt-hours (GWh) in 2025 and significantly higher when demand is compared with nameplate (nominal) capacity. This is primarily due to major producers in Asia, which together manufacture more than 75 percent of all batteries sold globally.⁴ The region's leading position, especially in the upstream and midstream supply chains, has led to imbalances across the global battery value chain (Exhibit 1). Governments have designed incentives to build up more localized industries, and the value of these policies is significant. But for globally competitive battery manufacturing industries to emerge outside of Asia over the next ten years, companies will need to do far more than ensure regulatory compliance. Challenges will need to be overcome on multiple fronts spanning supply chains, talent management, operations and technology. McKinsey Battery Insights expects a global battery market size of 4.2 TWh in 2030 and 6.8 TWh by 2035, with more than 85 percent of this demand driven by Li-ion batteries, primarily due to the strong demand for battery electric vehicles (EVs) and energy storage. This 2030 demand projection is lower than our [2023 estimate](#) of 4.7 TWh.⁵ but still well above a 2019 estimate of 2.6 TWh. The comparison underscores how quickly the industry has expanded and evolved and a possible inflection point for global growth. But without structural cost and yield improvements, producers could see constrained capacity growth in the years to come.

Asia's leading position, especially in the upstream and midstream supply chains, has led to imbalances across the global battery value chain.

¹Teo Lombardo et al., "The battery industry has entered a new phase," IEA, March 5, 2025; "Battery demand to reach 1.6 TWh in 2025 as Benchmark Week comes to a close in LA," Benchmark, November 21, 2025.

²"Global energy storage growth upheld by new markets," BloombergNEF, June 18, 2025.

³"Lithium-ion battery pack prices fall to \$108 per kilowatt-hour, despite rising metal prices," BloombergNEF, December 9, 2025.

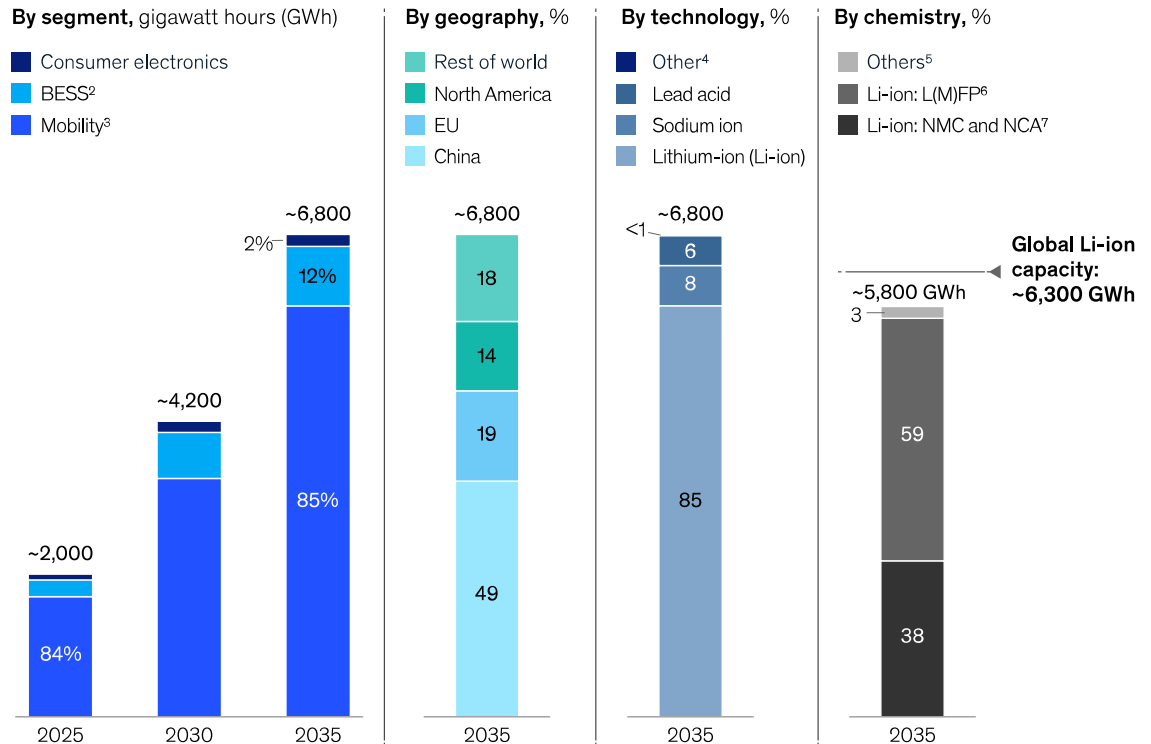
⁴Teo Lombardo et al., "The battery industry has entered a new phase," IEA, March 5, 2025.

⁵"[Battery 2030: Resilient, sustainable, and circular](#)," McKinsey, January 16, 2023; *A vision for a sustainable battery value chain in 2030: Unlocking the full potential to power sustainable development and climate change mitigation*, World Economic Forum, September 2019.

Exhibit 1

The battery industry can be broken down into different segments, geographies, technologies, and chemistries.

Global battery demand and market split, base view¹ (Q3 2025 update)



¹Balanced forecast in our global battery demand projection.

²Battery energy storage systems.

³Includes passenger cars, commercial vehicles, 2- and 3-wheelers, off-highway, railway, aviation, and maritime.

⁴Other technologies include nickel-metal hydride, redox flow, etc.

⁵Includes lithium cobalt oxide, lithium manganese oxide battery, lithium sulfur, solid-state batteries, etc.

⁶Lithium iron phosphate and lithium manganese iron phosphate.

⁷Lithium nickel manganese cobalt oxide and nickel cobalt aluminum.

Source: McKinsey Battery Insights; McKinsey Center for Future Mobility

McKinsey & Company

Technology road map to 2035

Battery technologies affecting all cell components have seen remarkable advances in recent years. Not unlike the semiconductor industry and the computing power of chips, lithium iron phosphate (LFP) and lithium nickel manganese cobalt oxide (NMC) battery chemistries have seen steady evolution rather than disruption. Commercial lithium-ion (Li-ion) battery energy density, for example, has more than doubled in the past 15 years. Expect Moore's Law to apply to batteries through 2035 as technologies and manufacturing processes continue to evolve (exhibit).

For cathodes, LFP is likely to remain the cost leader due to lower-cost material ingredients. With its lower cost-per-kWh and higher thermal stability, it will continue anchoring the mass-market electric-vehicle (EV) and battery energy storage system (BESS) markets. The lithium manganese iron phosphate variant (manganese increases energy density) is a promising evolution for slightly longer-

range vehicles and more compact packs. We forecast LFP's market share to reach 49 percent in 2030 and 52 percent in 2035 in the European Union, with the North American share reaching 47 percent in 2030 and 49 percent in 2035.

NMC, which also now encompasses high-manganese variants, will likely remain the performance leader for long-range and premium EVs. This chemistry does face heightened price and sourcing risks relative to nickel, cobalt, and manganese. We forecast NMC's market share to reach 46 percent in 2030 and 43 percent in 2035 in the European Union, with North American share reaching 46 percent in 2030 and 45 percent in 2035.

For anodes, silicon blends in graphite anodes will likely continue to trend higher. Battery makers are combining graphite (natural or synthetic) with silicon blends to increase energy density and charging performance without compromising durability and lifespan. We estimate that the typical silicon percentage in anodes will increase from under 5 percent in 2025 to approximately 10 percent by 2035. Because most graphite processing occurs in Asia, US producers will need to address an acute FEOC-related imbalances in the anode supply chain.

Electrolyte and separator technologies will likely continue to improve. Firms are designing next-generation electrolytes to handle higher voltages and cold-temperature performance. And they are creating separators with a thin ceramic layer to make cells safer and more durable under stress or heat.

There are some challengers to conventional Li-ion batteries, such as sodium-ion and solid-state batteries. Sodium-ion batteries are a promising alternative to lithium-based chemistries for small EVs and BESS due in part to reduced supply chain risks. No lithium, nickel or cobalt is required for these batteries. Leading players had been targeting mass production for EVs by 2026, but expectations have cooled as LFP prices have dropped. To be competitive in the years to come, sodium-ion batteries will have to match LFP's performance, cost, and durability.

Semi-solid-state batteries are maturing toward mass production. Commercial scale production of all-solid-state batteries, however, is unlikely until after 2030. The success of this alternative is contingent on overcoming manufacturing hurdles (such as achieving good interfaces and production yields) to establish competitive performance, longevity, and cost. We estimate that by 2030, solid-state battery energy density will be 400 to 450 watt-hours per kilogram, and production cost will be approximately \$100 per kilowatt-hour. We also project durability of 1,000 to 1,500 cycles when the battery is operated between 5 percent and 95 percent state of charge, and charging and discharging speed capability of 12 to 20 minutes.

Barring unforeseen breakthroughs, it will become harder for technological challengers to dethrone conventional Li-ion batteries as EV and BESS markets mature. If existing solutions are adequate—meaning EVs have sufficient range, and energy storage solutions become standardized—the barriers to change will only become greater.

We also anticipate step changes in the manufacturing process. Battery factories are likely to become significantly more efficient and productive in the coming years. We expect dry electrode coating to reduce capital expenditures and energy use, high-speed stacking to improve throughput and precision, and advanced battery-formation approaches to save time and improve cell consistency. Greater integration of digital twins and inline analytics can improve quality control. Taken altogether, these manufacturing advances have the potential to improve yields up to as much as 95 percent.

Exhibit

Battery technologies and manufacturing processes are likely to continue evolving through 2035.

	2010s		2020s		2030+
Lithium-ion cathode active material	Lithium cobalt oxide	Lithium NMC ¹ and NCA ² Lithium iron phosphate (LFP) Lithium manganese oxide	NMC and NCA LFP	NMC and NCA LFP and lithium manganese iron phosphate (LMFP)	NMC and NCA LFP and LMFP Lithium manganese-rich
Separators and electrolytes	Polymer and liquid	Polymer and liquid	Polymer and liquid Semi-solid	Polymer and liquid Semi-solid	Polymer and liquid Semi-solid Solid
Lithium-ion anode active material	Graphite	Graphite	Graphite	Graphite Graphite and silicon	Graphite-based Silicon anode Lithium metal
Casings	Cylindrical	Cylindrical Pouch	Prismatic Cylindrical Pouch	Prismatic Cylindrical Pouch	Prismatic Cylindrical Pouch
Alternative technologies	Lead acid	Lead acid	Lead acid	Lead acid Sodium ion	Lead acid Sodium ion Lithium sulfur

¹Nickel manganese cobalt oxide.

²Nickel cobalt aluminum.

Source: McKinsey Battery Insights

McKinsey & Company

This article focuses on the Li-ion battery industry and offers industry and public stakeholders in the United States and Europe (including Asia-based firms operating there) a fact base for building competitive global battery supply chains and unlocking the value of emerging technologies. By 2035, expect industry winners in various regions to excel in four areas:

1. They will achieve cost leadership at scale through a mix of optimizing bills of materials, improving manufacturing benchmarks, and other actions.
2. They will master industrialization excellence, operating efficiencies, and rapid ramp-up to achieve high yields.
3. They will build regionalized supply chains, including manufacturing equipment. Winners in the European Union and the United States will take advantage of policy incentives in their respective regions.
4. They will deliver credible technology road maps with respect to lithium iron phosphate (LFP), lithium nickel manganese cobalt oxide (NMC), and solid-state batteries, as well as manufacturing processes such as dry electrode coating (see sidebar, “Technology road maps to 2035”). Winners will bring new technologies to market faster than competitors.

Key to survival: Cost efficiency

In 2025, Li-ion battery pack prices fell by 8 percent, maintaining their downward trajectory after a 20 percent reduction in 2024.⁶ This drop was a clear warning to pack producers: Competition and margins are tightening as the global battery supply significantly exceeds demand.

Going forward, costs are expected to remain relatively stable and possibly even rise slightly as continued technological improvements are balanced by a gradual rebound in material prices. Based on a low-price scenario for raw materials, we project global average per-kWh pack costs of \$75 to \$90 for NMC and \$55 to \$65 for LFP by 2035. A price floor is not guaranteed, although critical mineral cycles and policy constraints affecting supply chains could reverse declines.

The key to survival, therefore, is cost efficiency. To mitigate the impacts of sustained low prices and potential market volatility, we suggest battery cell makers and OEMs consider a cost-efficiency checklist:

1. Bill of materials plan. More cost-efficient battery production will require adjusting LFP and NMC pack manufacturing capacity in response to shifting market demands. Producers should also closely track raw material availability and related cost increases. A 50 percent rise in lithium and nickel prices, for example, would likely push cell costs 10 percent higher, while a similar increase in raw graphite and manganese prices would have a much smaller cost effect. However, producers can partially offset price exposure by optimizing cathode loading and

⁶ “Lithium-ion battery pack prices fall to \$108 per kilowatt-hour, despite rising metal prices,” BloombergNEF, December 9, 2025.

silicon-anode blending into cells to improve energy density and charging speed. The bottom line: Cost resilience increasingly depends on optimizing the bill of materials, rather than commodity prices.

2. Manufacturing cost curve. Battery producers will need to improve multiple manufacturing benchmarks. These include production line productivity (annual GWh output per line), energy intensity (input per kWh produced) and kWh-weighted capital expenditures, as well as labor productivity. We suggest the following target benchmarks:

- production line productivity of 80 to 90 percent after reaching the steady state
- €50 million to €80 million per GWh in capital expenditures, depending on produced battery chemistry
- twenty to 35 kWh energy input per kWh of capacity produced, depending on produced battery chemistry
- thirty to 40 full-time employees per GWh

3. Yield and scrap rates. A battery factory's yield and scrap rates are crucial profit swing factors. As a production line matures, producers typically see the percentage of usable battery cells produced rise while scrap rates fall. Scrap rates during a facility's early ramp-up period can reach 70 or 80 percent, and disposal of hazardous waste can be costly. The ability to increase a line's first-pass yield and accelerate battery cell formation and testing is a major profit lever. This is partly because cost savings achieved through scrap reduction usually exceed any savings realized by negotiating lower raw material prices.

4. Localization pros and cons. Producers in the United States and European Union should assess the cost advantages (or lack thereof) of compliance with the Inflation Reduction Act (IRA) and the Critical Raw Materials Act (CRMA), respectively. In some cases, a China-sourced bill of materials may still offer cost advantages. Producers should base localization decisions on regional cost-benefit analyses, such as comparing the total cost of a US-produced LFP cell with an imported Chinese equivalent (Exhibit 2).

Operational excellence is a durable advantage

Anticipating sharply rising demand for Li-ion batteries, manufacturers in recent years have pushed up global production capacity.⁷ That trend will continue as more gigafactories (especially in Europe and North America) come online; nearly 700 GWh in additional capacity was under construction in just the United States as of early 2025.⁸ We expect global plant nameplate capacity to reach about 6.7 TWh by 2030. That represents a 67 percent increase over a current estimated capacity of four TWh.

⁷ "Lithium-ion battery pack prices see largest drop since 2017, falling to \$115 per kilowatt-hour," BloombergNEF, December 10, 2024.

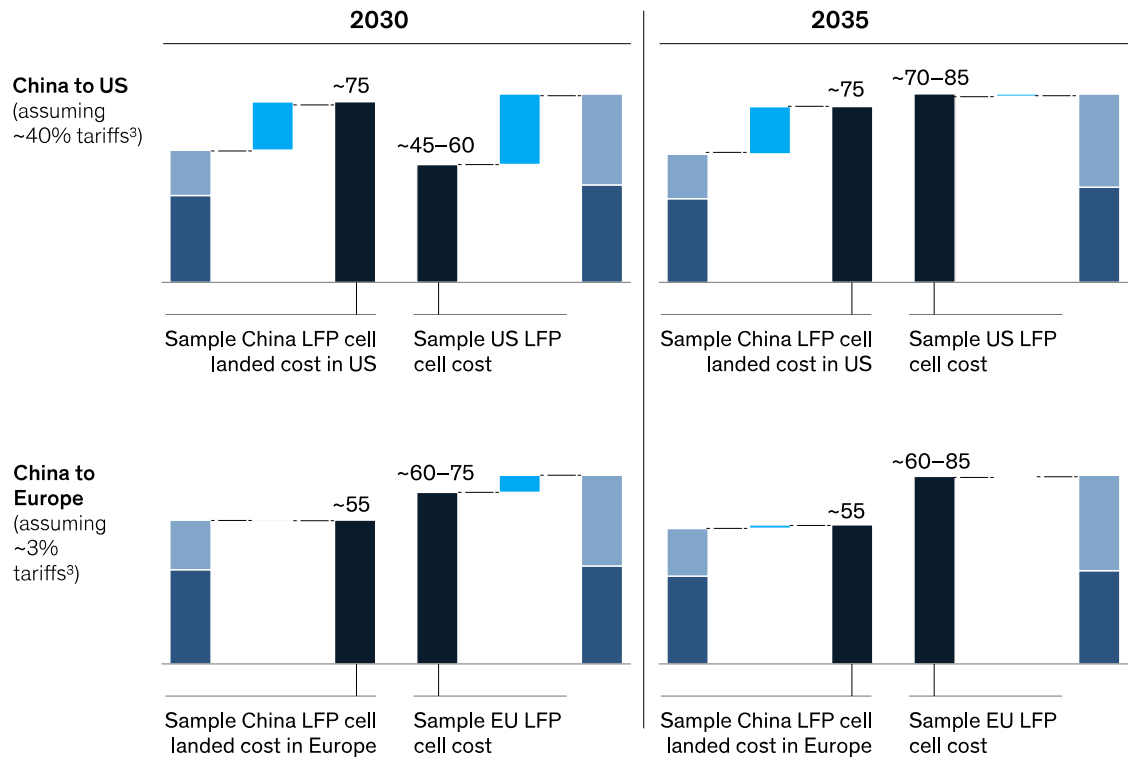
⁸ Teo Lombardo et al., "The battery industry has entered a new phase," IEA, March 5, 2025.

Exhibit 2

A region-by-region cost analysis can help battery producers make localization decisions.

Comparison of lithium iron phosphate (LFP) battery cell costs,¹ \$/kWh, 2025 real

■ Bill of materials² ■ Transformation cost ■ Tariffs or subsidies



¹All cost values are in 2025 real terms (inflation-adjusted).

²Including transportation costs.

³Based on current tariffs on China battery cell imports in 2025.

Source: McKinsey Battery Insights

McKinsey & Company

Yet ramping up facility output to reach targeted nameplate capacities remains a frequent bottleneck. Despite current global overcapacity and demand exceeding one TWh in 2024, many production lines underperform. Delays in achieving high overall equipment effectiveness (OEE) and stable production quality threaten producers' ability to compete in an increasingly crowded global market.

In this environment, a producer's ability to achieve operational excellence creates durable comparative advantage. The ability to accelerate ramp-up time to a 95 percent yield rate, for example, will set industry leaders apart. In our view, the playbook for reducing or eliminating battery production ramp-up pain and sustaining high productivity has three components.

1. The OEE triangle. For battery factories, the three sides of the OEE triangle are availability (no unplanned downtime), performance (optimal throughput) and cell production quality (high yield). Producers should target best-in-class OEE goals during ramp-up, reaching yields above 95 percent and OEE of at least 85 percent within 18 months after the first cell is produced.

2. Process innovations. To compete successfully, manufacturers must integrate new process innovations into their operations at scale. These innovations, which include dry-electrode coating, high-speed stacking, and advanced formation protocols, can reduce cell production costs through energy and capital expenditure savings. We estimate that dry-electrode coating in Europe, for example, can save €2 to €3 per kWh in costs by reducing both energy consumption and capital expenditure requirements.

3. People and systems. In battery factories, production process stability and product quality are tightly linked. Operators' skills are crucial for bolstering productivity and ensuring quality consistency. The ability to attract the right technical talent can be decisive, and many battery start-ups have failed due to skills deficits. Producers should define skill models and prioritize structured systems for training and evaluating factory worker competencies. Real-time production line monitoring and inspection (inline metrology), manufacturing execution systems, and statistical process control (SPC) tools for critical steps—such as slurry, calendaring density, and electrolyte fill—can all improve cell productivity and quality.

Two regions need to step up: Europe and the United States

Asian producers are excelling today, but momentum to localize the battery industry continues to grow in Europe and North America. Despite having higher labor, electricity, and raw material costs, these two regions are working to develop globally competitive battery value chains, supported by policies designed to encourage growth.

Europe and the United States have similar ambitions to mature their respective battery industries, although both areas have distinct policy approaches. Through its Green Deal Industrial Plan and related legislation, the European Union is now pushing to expand its manufacturing capabilities to meet at least 40 percent of the region's clean technology needs by 2030. Specific domestic targets for that year extend to strategic raw materials: The European Union aims to mine 10 percent, process 40 percent, and recycle 25 percent of these materials by 2030.⁹ While it has taken tangible steps to spur progress, including selecting 47 strategic projects aligned to the CRMA to boost capacities,¹⁰ funding and permitting gaps persist. The current ongoing strategy dialogue between the European Commission and the automotive industry is also discussing a new form of EV ramp-up support that will include new subsidies across the battery value chain.

⁹ "European Critical Raw Materials Act," European Commission, accessed December 9, 2025.

¹⁰ "Commission selects 47 Strategic Projects to secure and diversify access to raw materials in the EU," European Commission, March 24, 2025.

In the United States, the IRA has spurred major investments in domestic battery production. While the federal \$7,500 tax credit for EVs expired in September 2025,¹¹ almost certainly affecting EV sales, the IRA's crucial Section 45X tax credit remains. That \$35 per kWh credit effectively lowers the cost of battery cell manufacturing by 30 to 50 percent, according to McKinsey analysis. This incentive can offset operating expenses and yield losses during production ramp-up, helping US domestic producers narrow cost gaps with China.

Current regionally imbalanced supply chains present potential volatility (Exhibit 3). For example, both the United States and the European Union claim less than 10 percent of the global share of anode graphite and cathode precursor production. This midstream processing imbalance poses a challenge for maturing industries in both regions. If the United States and the European Union want to scale globally competitive battery value chains, this would require reshoring cell assembly as well as mineral refining and processing (Exhibit 4).

Success factors in Europe

Establishing a resilient, competitive battery value chain in Europe will likely require sustained public–private alignment. Despite the region's ambitions, much previously announced production capacity has not materialized, and most current production is from non-European companies. This underscores the need for a coordinated strategy to build competitive scale and capabilities across the value chain.

We estimate that for the continent's battery industry to reach global competitiveness, €200 billion to €300 billion will need to be invested by 2035. Capital will need to be deployed across the full value chain—from raw materials to recycling—through a game plan approach. Specifically, control of key areas, such as refining, cell manufacturing and recycling, is essential to build resilience and capture more value within Europe.

Demand for batteries in the region is projected to continue rising sharply, and not only for EVs. Growing wind and solar energy production is driving a BESS boom in Europe, with installed capacity expected to grow to 60 GWh by 2030, up from 20 GWh in 2024.

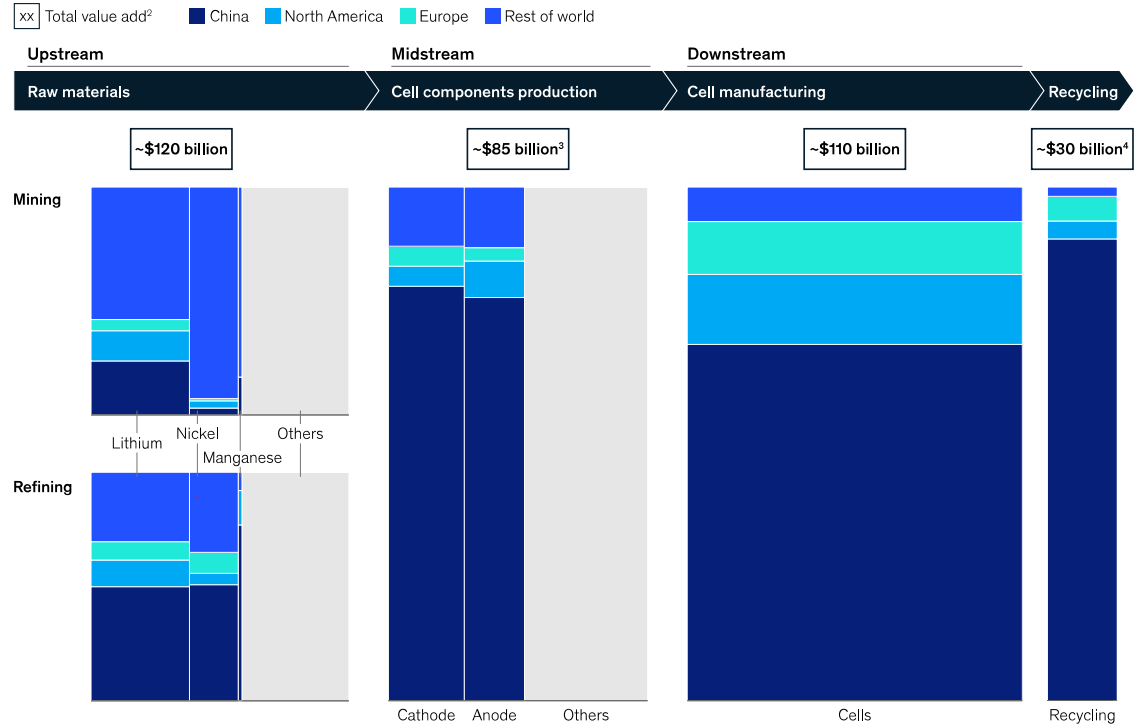
If costs and permitting align, the opportunity for a localized battery supply is clear. The path to creating a competitive value chain in Europe is less clear, but success is far more likely if policymakers encourage supply chain participation and R&D investment by creating attractive frameworks to draw in leading global industry players and enable technology localization. Another crucial success factor is enabling skill development and technology transfers through strategic partnerships and joint ventures. The region could close industrial capability gaps by offering joint-venture structures eligible for government incentives that can bring leading battery industry players together with European producers. Facilitating public–private action through faster permitting, access to financing and cross-border cooperation is also crucial. In particular, streamlining approval procedures to accelerate production facility development and supply chain localization in alignment with CRMA goals is key.

¹¹ "Clean vehicle tax credits," Internal Revenue Service, updated November 26, 2025.

Exhibit 3

The majority of the global battery supply chain is concentrated in China.

Share of core steps along lithium-ion (Li-ion) value chain by project location, 2035 (estimated),¹ %



Note: The width of each bar reflects the relative value add of each step in a battery cell.

¹Based on currently announced projects.

²Based on the value add in each value chain step for conventional Li-ion batteries (not including next-gen batteries).

³Including all battery components (cathode, anode, electrolyte, separator, current collector, etc.).

⁴Total market revenue is based on the value of recovered materials, estimated using feedstock volume, collection rates, recycling yields, and prevailing metal prices.

Source: McKinsey Battery Insights; McKinsey MineSpans.

McKinsey & Company

While full localization is not necessary, the battery industry's economic impact in Europe could be bolstered if, for example, a higher share of material processing occurs within the region. More broadly, only a pan-European, partnership-driven approach can establish a competitive battery value chain able to secure Europe's transition to EVs and growing renewable energy production.

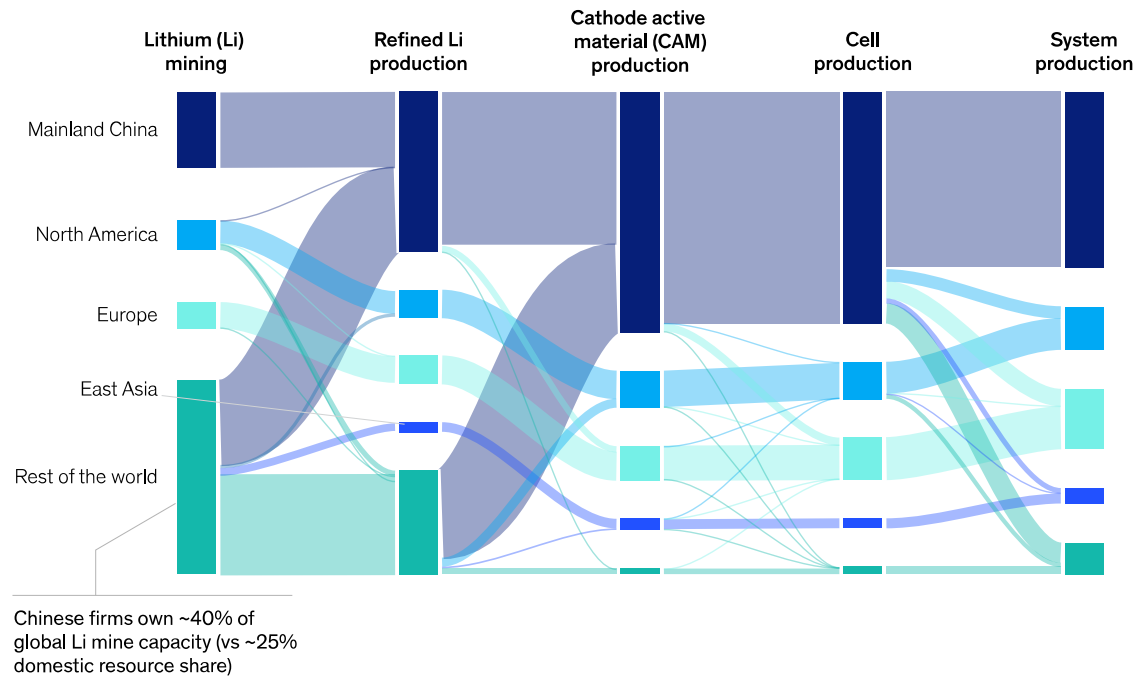
Success factors in the United States

Broadly speaking, US efforts spurring its battery industry forward mirror Europe's ambition and the European Union's incentive-based approach. But the country operates within a distinct

Exhibit 4

A 2035 scenario shows global material flows based on announced supply capacities, estimated regional demand, and defined optimization rules.

2035 illustrative global flows across the battery value chain based on production locations¹



¹This exhibit illustrates one possible future scenario of global material flows based on announced supply capacities, estimated regional demand, and a defined set of optimization rules. Actual flows may differ, and alternative scenarios can be explored by adjusting key underlying assumptions. Source: McKinsey Battery Insights; McKinsey MineSpans; McKinsey Center for Future Mobility

McKinsey & Company

policy and market context. Although the US Congress did rollback clean-energy-related credits and other policies in 2025, the IRA has nonetheless led to additional large investments in the domestic battery sector.

As in Europe, rapid growth in BESS is becoming an increasingly important driver of US battery demand, alongside continued EV uptake. In 2024, energy storage installations reached 37.1 GWh.¹² Installations were projected to exceed 50 GWh in 2025, setting a new record. While the One Big Beautiful Bill Act passed by Congress in July 2025 largely left existing battery storage

¹²Brian Martucci, "US deploys record energy storage in 2024, but Trump policies cloud outlook: WoodMac/ACP," Utility Dive, March 21, 2025; Brian Martucci, "US sees record utility-scale storage deployments, but dropoff looms: report," Utility Dive, September 30, 2025.

tax credits intact, the legislation did enact more stringent foreign entity of concern (FEOC) material requirements, tightening localization rules.¹³

Regulatory stability, including policy incentives, is an important factor shaping the future of the US battery industry. Other key enablers include a range of federal and state-level incentives supporting emerging industrial clusters in the Southeast and Midwest known as the “Battery Belt.”¹⁴ Success factors also include public–private partnerships to derisk large-scale projects and R&D support and workforce programs to build technical capacity.

Ultimately, the United States could emerge as a self-sustaining and globally competitive battery hub. If that happens, localized ecosystems in part cultivated by government policy and investments will likely be a decisive force.

What it would take to build a regionalized, end-to-end supply chain

As demand for batteries continues to rise, Europe and the United States are trying to reduce their supply dependency on Asia. Even with all stakeholders aligned, ramping up localized production to meet demand by 2030 without significant reliance on supply chains in Asia will be challenging.

In our view, there are four requirements for successfully building regionalized, competitive and resilient end-to-end supply chains:

1. Establish an incentive price

Given the scale and maturity of the battery supply chain in Asia, if the goal is to scale localized supply chains in the United States and Europe, these regions should consider new ways of incentivizing market participants and derisking capital. To attract the right participants, governments and industry will need to create clearer price signals and incentive mechanisms that are technology-agnostic, player-agnostic, and focused on the supply chain end to end. This requires insights into battery ecosystem dynamics as well as the specific challenges at each part of the value chain. The first step is understanding the techno-economics of each part of the chain, including the technology pathways, player landscape, cost drivers, and capital investment and returns. Strategic considerations involve understanding how to attract the right participants and capital by creating the right derisking and incentive mechanisms, factoring in longer-term regional price signals, risk-adjusted returns, contract terms, geopolitical exposure, and offtake certainty.

A forthcoming article will take a deeper look at how to design effective incentive mechanisms for the battery value chain to enable and scale battery supply chains in North America and Europe.

2. Map foreign exposure and establish digital traceability

In terms of strategy and governance, US producers can map and manage their exposure to FEOCs across the entire value chain—from mining and refining materials to cell component, cell,

¹³ Gracelin Baskaran and Meredith Schwartz, “Impacts of the One Big Beautiful Bill Act on the mining sector,” Center for Strategic & International Studies, July 9, 2025.

¹⁴ Jay Price, “Growing ‘battery belt’ for EV plants could spark economy,” NPR, April 7, 2024; Rebecca Bellan, “Tracking the EV battery factory construction boom across North America,” Tech Crunch, February 6, 2025.

and pack production. This requires establishing governance frameworks that monitor equity ownership, board control, intellectual property rights, and tolling arrangements at every stage. Manufacturers should implement digital mapping tools to assess FEOC-related risks, supported by legal and compliance reviews. Designing compliant partnership structures will be essential to maintain credit eligibility under US government rules while preserving flexibility in sourcing.

Similarly, digital traceability can verify material provenance and ensure compliance with both IRA and CRMA standards. Producers should consider implementing data management and evaluation systems (such as part-level mass-balance tracking and supplier scorecards) to automatically verify that the critical minerals they use meet domestic-content requirements. This can enhance auditability and credit eligibility.

3. Ramp up mining and refining

Localizing the mining and refining of key materials such as lithium, nickel, manganese, and graphite is essential to meeting domestic-content requirements and ensuring supply security. Expanding domestic and allied-region capacity (through CRMA-designated strategic projects in the European Union and Department of Energy-supported initiatives in the United States) is vital for compliance with emerging ownership and sourcing rules. There is progress on this front: of the European Union's 47 strategic projects aligned with CRMA goals, 30 are related to key battery raw materials such as lithium and graphite. As trade and industrial policies continue to evolve, increasing local content will be key.

4. Build midstream capacity

Building midstream capacity is critical to closing regional supply gaps. Strengthening it helps capture value, meet battery performance standards and regulatory requirements, and ensure market agility. Localized investments should prioritize four things:

- synchronizing cathode and anode active material production with battery cell start-of-production timelines
- solvent recovery systems
- alternatives to polyvinylidene fluoride (a plastic polymer used as a binder)
- recycling black mass and active material

As producers look to expand regional midstream activities, they should view recycling as a lever for supplementing local upstream battery materials, beyond ensuring compliance with emerging regional recycled-content requirements. Ultimately, recycled content can help close significant capacity gaps in the European Union and North America.

Beyond mobility: BESS hits its stride

BESS is arguably the world's fastest-growing clean-energy technology. Even amid policy uncertainty and shifting incentives across markets, the global volume of utility-scale BESS installations keeps setting records.¹⁵ McKinsey estimates that total installed BESS capacity will reach 200 GWh by the end of 2025 and will likely reach between 500 and 700 GWh by 2030.

The growing share of renewables in power grids is driving the rapid rise of BESS. As wind and solar production grows, so does the need for flexibility such as storage that can smooth intermittency and balance grids. In short, BESS is fast becoming a foundational component of the world's energy infrastructure.

Business case about more than chemistry

While battery chemistry affects cost and performance, BESS project economics are more dependent on other variables. These include the costs of project design, construction, interconnection, degradation and augmentation strategy, and (most important) the different revenue streams such as ancillary services, energy arbitrage, and capacity payments, which vary by region and can evolve over time for a single project.

Spotlight: BESS in North America and Europe

In the United States, 2025 is expected to be another record year for BESS additions.¹⁶ Surging data center power demand due to growth in AI and cloud computing is catalyzing BESS installations, which can enhance power reliability and reduce grid strain. Growth is also being powered by IRA incentives and independent system operator market reforms that reward capacity and nodal arbitrage, driving projects near major solar hubs in states including Texas and California. On the regulatory and compliance front, supply chain and content origin rules are reshaping procurement. FEOC restrictions now apply to federal grid-storage tax credits,¹⁷ compelling BESS project developers to trace battery materials and select compliant vendors accordingly, also spurring investment in local cell manufacturing dedicated to BESS.

In Europe, the current BESS project pipeline—amounting to about 60 GWh by 2030—lags well behind the storage volumes needed to balance renewable growth. Uptake depends on factors including revenue clarity relative to locational signals and ancillary services, as well as permitting speeds. Without accelerated permitting and clarity on prices, the risk of a large storage gap in Europe that hinders the region's grid-balancing capacity is significant.

Technology outlook

By 2035, the BESS market is likely to segment by use-case and duration. LFP will remain the workhorse for up to six-hour systems due to the technology's cost, safety and reliability. Once supply chains mature, sodium-ion batteries may offer a low-cost alternative to Li-ion batteries for stationary storage. Meanwhile, flow and other long-duration chemistries will remain niche until they are proven at scale and reach total cost of ownership crossover with Li-ion batteries for multi-hour use.

¹⁵ "Global energy storage growth upheld by new markets," BloombergNEF, June 18, 2025.

¹⁶ "U.S. battery capacity increased 66% in 2024," US Energy Information Administration, March 12, 2025.

¹⁷ Keith Martin et al, "Working through the FEOC maze," North Rose Fulbright, July 8, 2025.

Priorities for the next 24 months: Five actions for EU and US leaders

As global demand pushes higher and both technologies and regulatory approaches evolve, battery producers find themselves navigating growing complexity. We recommend leaders in the European Union and the United States prioritize taking swift action in five areas to build competitiveness:

Lock in cost leadership. As margin pressure rises and raw material volatility persists, leaders will need to closely review materials and operations for value, striking the right balance between cost and performance. Manufacturers should run bill-of-materials optimization waves relative to both LFP and NMC to balance cost and performance. Tightening supplier contracts and rebidding key inputs, while also reducing scrap and improving yields, are also crucial steps to ensure cost leadership.

Industrialize fast. As new factories ramp up, leaders should focus operations teams on mitigating all risks to start-of-production (SOP) timelines. Specific tactics include defining golden-batch standards, validating process capability (measurement system analysis), and establishing SPC libraries for early fault detection. Capital expenditure phasing should align to OEE gates, ensuring that each investment meets productivity milestones. Leaders in both regions will be able to deliver new factories with a faster time to SOP.

Balance derisking tariff exposure with maximizing policy support. Between 2025 and 2027, the EU Innovation Fund is providing more than €3 billion in financial incentives for battery manufacturing, with the EU Commission releasing the automotive package, including a battery booster, in December 2025. At the same time, with Prohibited Foreign Entity rules and CRMA targets in place in the United States and the European Union, respectively, producers should consider implementing compliance programs. This may involve prioritizing joint venture/ownership structure audits and establishing documentation processes.

Accelerate BESS deployment. Leaders of component and system manufacturers should focus on identifying the best value proposition for EU and US markets. In the United States, for example, this could mean the production of localized LFP cells or optimizing full systems to result in the lowest capital expenditures and operating costs. Localization of control and software components can be another lever in these regions.

Place smart tech bets. Three areas in particular merit investment. Consider piloting dry electrode coating lines, advancing silicon-blend anode integration, and maintaining the ability to pivot into sodium-ion batteries for stationary storage as supply chains mature. We also suggest keeping a close watch on solid-state technology developments (especially in the 2027–30 period) to ensure readiness to implement.

Taken together, these five moves position leaders for the future by locking in cost advantages, scaling production efficiency, leveraging innovation and mitigating both policy and supply risks.

Find more content like this on the
McKinsey Insights App



Scan • Download • Personalize



Andreas Breiter is a partner in McKinsey's Bay Area office, **Denny Thiemig** is a partner in the Berlin office, **Patrick Schaufuss** is a partner in the Munich office, **Raphael Rettig** is a partner in the Dusseldorf office, **Wenting Gao** is a partner in the Houston office, and **Yunjing Kinzel** is an asset leader in the Frankfurt office.

The authors wish to thank Alexandre van de Rijt, Jakob Fleischmann, Martin Linder, Mikael Hanicke, Shan Xiong, and the broader Battery & Energy Storage Insights and MineSpans teams for their contributions to this article.

Copyright © 2026 McKinsey & Company. All rights reserved.